Bruno Henrique Nunes

Applied AI / Machine Learning Engineer

Brazil | Open to remote opportunities hiring from Brazil

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Professional Summary

AI and software engineer with a completed MSc in Artificial Intelligence from UNICAMP. Build multi-agent applications, chatbots and RAG workflows at IBM, supported by research in computer vision and reproducible machine learning. Bring enterprise experience in Python/SQL workflows, API integration, testing and international delivery.

Technical Skills

Applied AI and ML: Python, PyTorch, multi-agent workflows, chatbots, RAG, IBM watsonx.

Computer vision: Deep learning, CNNs, Transformers, semantic segmentation, model evaluation, ensemble inference and reproducible research.

Data: SQL, data preparation, quality control, patient-level splits and provenance.

Software delivery: FastAPI, REST APIs, AWS Lambda, AWS SES, Git, Azure DevOps CI/CD.

Professional Experience

Application Developer — IBM Aug 2026 – Present

- Build multi-agent harnesses and applied-AI application components; explore datasets and train machine learning models for data-science use cases.
- Architected and developed a production-deployed SDLC Run Insights Chatbot in Python/FastAPI, retrieving run evidence and platform knowledge and using IBM watsonx LLMs for grounded responses.
- Implemented prompt-injection and output-safety guardrails, observability, automated tests and production deployment support for the chatbot.

Selected Research and Projects

Deep Learning for Prostate Histopathology Feb 2024 – Sep 2026

Master's research, University of Campinas (UNICAMP)

- Developed a research pipeline for binary segmentation of prostate cancer in histopathological images, using CNN/Transformer-based modeling and PyTorch.
- Implemented data preparation and quality control, patient-level data partitioning, model training, ensemble optimization and final evaluation.
- Preserved configurations, data provenance and experimental outputs to support reproducibility. The REDU deposit contains generated research artifacts; original DIAGSET images are not redistributed.

Code: github.com/bhnunes/Master_Coding | Artifacts: doi.org/10.25824/redu/EDPCF2

GEODITE — Equivariant Interatomic Potentials

Contributed to model implementation, training and evaluation for an equivariant message-passing model for interatomic potentials.

Co-author, *Equivariant Interatomic Potentials without Tensor Products* (arXiv preprint, January 2026).

arxiv.org/abs/2601.15492

Teaching Experience

Teaching Assistant — UNICAMP Jul 2024 – Dec 2025

Supported more than 200 undergraduate students in database management and object-oriented programming; led problem-solving sessions in SQL, relational algebra and OOP and supported assessment. Internship appointment.

Selected Earlier Experience

Software Developer — PAteam (freelance) Jan 2024 – May 2024

Developed Python automation and data-ingestion workflows, including AWS Lambda-hosted extraction, REST API integration and AWS SES messaging.

Software Developer — Expleo Group (contract) May 2022 – Aug 2023

- Built automation and API integrations for a legacy-to-Salesforce data migration using Python, Robocorp, Salesforce REST APIs and SQL Server.
- Clarified requirements with client and delivery teams; documented workflows and supported deployment through Azure DevOps CI/CD, migration monitoring and issue resolution.

Senior RPA Developer — Nestlé Feb 2022 – Aug 2023

- Designed and implemented 13 enterprise automations using Python, Blue Prism, PowerShell, SQL and Automation Anywhere; worked with stakeholders in and outside Brazil.
- Supported and reviewed code across more than 10 projects, mentored developers, and managed change requests for release and approval of more than 50 robots.

Senior RPA Developer — Deloitte Nov 2021 – Feb 2022

Translated stakeholder requirements into Python, Node.js and UiPath automation for a healthcare IoT integration project; presented solutions to clients and suppliers.

IBM — Earlier roles

RPA Developer: Oct 2019 – Nov 2021 | Software Developer: Oct 2018 – Oct 2019

IT Analyst: Jan 2017 – Oct 2018 | Support Specialist: Jan 2012 – Oct 2014

- Delivered global and regional automations with Python/Selenium, APIs, SAP, Db2, Cloudant and Blue Prism; built ad hoc tools with Python, VBA and JavaScript.
- As IT Analyst, created invoice analytics with VBA and Power Query, consolidating more than 20,000 records into over 10 validation KPIs; reported processing time fell from over three hours to about six minutes.
- As Support Specialist, provided English-language support to US and Canadian clients and used SQL for compliance checks and billing-adjustment reporting.

Education

MSc in Artificial Intelligence — University of Campinas (UNICAMP) 2024 – 2026

Completed September 2026; research in deep learning for prostate histopathology.

MBA in Computer Software Engineering — FIA Business School 2022 – 2023

Bachelor's in Engineering Physics / Applied Physics — University of São Paulo 2006 – 2010

Selected Certifications

AWS Certified Generative AI Developer – Professional Oct 2026

AWS Certified Machine Learning Engineer – Associate Jul 2026

AWS Certified Developer – Associate Oct 2025

AWS Certified Solutions Architect – Associate Apr 2025

Languages

Portuguese: Native. **English:** Full professional proficiency.

TOEIC Listening and Reading: 890 (September 2025).

Spanish and Italian: Elementary proficiency.